

MIAE 215 Arduino Project

This “mini-project” is a follow up on Arduino lab #3. Please finish that lab and then answer the question given below. The following items should be submitted as a Moodle assignment:

- a) The *.ino file for your Arduino program (eg “program.ino”).
- b) High quality photos (1-2 in number) of your project apparatus (*.jpg, etc.).
- c) A brief video demonstrating your project in operation (*.mp4, etc.).

No report is required. However, your program should include C++ comments that briefly explain how the program works (roughly one comment every 3-5 lines of the program).

Each student should work on the project separately. Collaboration is not allowed.

The main grading criteria are:

- a) Technical merit -- how correct and general is your solution.
- b) Originality -- how original / innovative is your project, including the program and its organization. Collaborating or copying will have a very negative impact on this aspect. Novel approaches will have a positive effect. However, the topic / objective of your project doesn't have to be very original as long as the implementation / program represents your own individual work (i.e. it “originates” from you).

Project Question

Add a new sensor to your apparatus from Arduino lab #3. You can select any sensor (other than the light sensor) from the Arduino kit or some other source. Possible sensors include:

- a) Joystick
- b) Temperature sensor
- c) Ultrasonic sensor

Additional sensor suggestions along with their references are included in the file “Arduino_kit_info.rar”. Don't cut and paste the included examples in your solution though -- write your own version / modify the examples to suit your purpose.

Write a program that uses the new sensor combined with the servo motor actuator from Arduino lab #3. The servo should respond to the new sensor input, ideally to perform some useful / interesting task. For example, controlling the servo position / velocity using the joystick or moving the servo like a dial in response to temperature changes. The program should be organized in terms of functions if possible.

You can use additional sensors (including the light sensor), actuators, and components for the project, but it's not required. You can also add mechanical design components (cardboard or wood construction is acceptable) to make the project more interesting, but

it's not required. Keep in mind the grade is mainly for the Arduino program, so don't spend too much time on mechanical design unless it makes for a more interesting / original Arduino program (that you have time to finish).

If you have trouble deciding on a project topic, you can pick one of the topics given above or just move the servo in some manner in relation to the new sensor input. The main thing is you pick a topic in a reasonable period of time so you can start working on it.

Finally, the amount of work you put into the project shouldn't be much more than an assignment or lab. Keep in mind the percentage weighting for the project in the course. However, there will be an Arduino problem on the exam (worth about 10% of the exam), so make sure you understand the related lab / project material. Working on the project is the best way to study for the Arduino question on the exam.